

Michael Dickinson

US Citizen | (541) 513-6290 | michael.dickinson.r@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of British Columbia

Graduation Apr 2028

Computer Science and Honors Mathematics (Double Major) – GPA 4.3/4.33

Vancouver, BC

- Teaching Assistant for CPSC 110 (Systematic Programming Design)
- Relevant Coursework: Computer Vision, Data Structures and Algorithms, Honors Linear Algebra, Numerical Approximation and Discretization, Object Oriented Programming

TECHNICAL EXPERIENCE

UBC Uncrewed Aircraft Systems

Sep 2024 – Present

Software Team Lead

Vancouver, BC

- Led software team (**20+ members**) to **4th Place Nationally** at AEAC 2026 student drone competition
- Added **SLAM system (cuVSLAM)** from **RealSense** RGB-D feed for autonomous flight, integrating nvBlox for path planning
- Architected real-time target detection on a **Jetson Orin Nano** with **YOLO and ROS2**, using sliding-window consistency check to reach 89% accuracy on competition targets
- Enabled autonomous takeoff and monitoring with a **ROS2-MAVLink** command and telemetry link, connecting over **RFD900x** to CubePilot flight controller with integration testing via ArduPilot SITL

MUX Lab - University of British Columbia

May 2026 – Present

Undergraduate Researcher

Vancouver, BC

- Submitted **first-author paper** to ACM CHI 2027 on live 3D reconstruction pipeline with Gaussian Splatting
- Integrated **RTAB-Map SLAM system** localizing on reconstructed scene for **<5cm pose error** on Bonn dataset
- Developed **optical-flow** based motion segmentation system with ego-motion compensation using **PnP RANSAC** and YOLOE object filtering for live detections at 93% precision, 88% recall
- Parallelized motion-segmentation, SLAM, and YOLO using Python multithreading, increasing throughput from **7** → **15 fps** and avoiding CUDA OOMs
- Awarded \$10,395 Work Learn International Undergraduate Research Award (WLIURA)

Twenty Ideas

Apr 2023 – Jul 2024

Software Engineering Intern

Eugene, OR

- Developed computer vision pipeline using **OpenCV** and **PyTorch** for automated truck damage assessment from photos, reducing manual inspection time by 70% and enabling instant repair cost estimates
- Integrated **Terraform IaC** workflows into deployment pipeline, allowing for **AWS infrastructure deployments** in under 10 minutes, including plan/apply stages for multiple environments and audit trail + rollback capability
- Reduced QA turnaround times to **< 20 minutes/build**, enabling nightly releases via GitHub Actions **CI/CD** pipeline, integrating with AWS for infrastructure deployments on **AWS Fargate + ECS**

SheerID

Apr 2022 – Sep 2022

Data Science Intern

Portland, OR

- Leveraged **Selenium web scraping** to **locate over 10,000 students** and provide them with custom promotions – led to over 1000 conversions for clients such as **Google and AT&T**
- Utilized **Python Pandas** to scrub **Postgres database**, flagging 300+ redundant user records for removal

PROJECTS

BERT + Vision Transformer – Multimodal Suicide Prevention Research | *PyTorch, NumPy, OpenCV*

- Engineered novel self-attention architecture fusing BERT with Vision Transformer (ViT) for multimodal analysis (image + text), achieving **85% accuracy** in suicidal ideation detection across 5,000 posts
- Awarded **American Psychological Association Achievement Award** for research on suicide intervention

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, TypeScript

Frameworks: ROS2, PyTorch, Pandas, NumPy, OpenCV

Tools: Linux, Git, Docker, AWS, PostgreSQL, GitHub Actions CI/CD